Life Sciences und Facility Management
ICLS Institut für Computational Life Sciences

Assignments

Assignment 1: Development of an In-House Process in the Field of Life Sciences

In this assignment, the focus is on the creative and methodological development of an original process in the field of life sciences. You will model the selected process using a BPMN (Business Process Model and Notation) diagram, identify relevant parameters, and define key performance indicators (KPIs). Additionally, you should consider uncertainties and random variables that may influence the process. The objective is to develop a well-founded and practical process that is both theoretically and methodologically sound.

1. Process Description and Modeling

- Develop a process of your choice within the field of life sciences (e.g., a biotechnological production process, a diagnostic procedure, or a clinical protocol).
- Create a BPMN diagram to visualize the process, ensuring clarity and comprehensibility in each step.

2. Parameters and Key Performance Indicators (KPIs)

- Identify and describe all relevant parameters of the process, including input variables, influencing factors, and key process variables.
- Select and define key performance indicators (KPIs) that will be used to evaluate the process. Provide a detailed description of these KPIs and their significance.

3. Stochastic Elements

- Identify and describe stochastic elements in the process. Explain the presence of random variables or uncertainties and analyze their potential impact on the outcome.

4. Presentation

- Present your findings in a **10-minute presentation**. The presentation should clearly and concisely outline your process, the selected parameters, KPIs, and the stochastic elements involved in the simulation.